

Equilibria

A Level Chemistry

Reversible reactions and dynamic equilibrium

A **reversible reaction** 可逆反应 can go both ways. The **forward reaction** 正反应 makes products; the **reverse reaction** 逆反应 turns products back into reactants. We show this with the sign $\rightleftharpoons$.

In a **closed system** 封闭系统 (nothing enters or leaves), the reaction reaches a **dynamic equilibrium** 动态平衡 when:

- the rate of the forward reaction equals the rate of the reverse reaction.
- the concentrations of reactants and products stay constant.

It is called *dynamic* because both reactions are still happening—they just cancel out. A closed system is needed, or products would escape and equilibrium could never be reached.

Le Chatelier's principle

Le Chatelier's principle 勒夏特列原理 says: if you change a system at equilibrium, the position of **equilibrium** 平衡 moves to oppose (reduce) that change.

Change you make	Which way the equilibrium moves
increase concentration of a reactant	towards the products
increase pressure	towards the side with fewer gas molecules
increase temperature	towards the endothermic direction
add a catalyst 催化剂	no shift (it speeds up both ways equally)

A catalyst lets equilibrium be reached faster, but it does **not** change the position of equilibrium.

Equilibrium constants

For a reaction at equilibrium, the **equilibrium constant** 平衡常数 links the amounts of products and reactants. For the reaction $aA + bB \rightleftharpoons cC + dD$:

$$K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

where the square brackets mean concentration in mol dm^{-3} .

For reactions of gases, we use **partial pressure** 分压 instead of concentration. The partial pressure of a gas is the share of the total pressure that it provides. It is found from the **mole fraction** 摩尔分数 (the fraction of all the moles that are that gas):

$$\text{partial pressure} = \text{mole fraction} \times \text{total pressure}$$

The constant written with partial pressures is K_p .

Only **temperature** changes the value of K_c or K_p . Changing concentration or pressure, or adding a catalyst, shifts the position of equilibrium but leaves the constant unchanged.

The Haber and Contact processes

These two industrial processes are chosen by balancing yield, rate and cost using Le Chatelier's principle.

- the **Haber process** 哈伯法 makes ammonia: $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$ (exothermic). It uses about 450 °C, 200 atm and an iron catalyst. A low temperature would give more ammonia but too slowly, so a moderate temperature is a compromise.
- the **Contact process** 接触法 makes sulfur trioxide: $2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3$ (exothermic). It uses about 450 °C, near 1–2 atm and a vanadium(V) oxide catalyst.

Acids and bases: the Brønsted–Lowry theory

A **proton** 质子 is simply an H^+ ion. The Brønsted–Lowry theory defines acids and bases by what they do with protons:

- an **acid** 酸 is a **proton donor** 质子供体 (it gives away H^+).
- a **base** 碱 is a **proton acceptor** 质子受体 (it takes H^+). A base that dissolves in water is called an alkali.

Common acids you must know: hydrochloric acid (HCl), sulfuric acid (H_2SO_4), nitric acid (HNO_3) and ethanoic acid (CH_3COOH). Common alkalis: sodium hydroxide (NaOH), potassium hydroxide (KOH) and ammonia (NH_3).

Strong and weak acids and bases

This is about how fully an acid or base splits up in water —not how concentrated it is.

- a **strong acid** 强酸 or **strong base** 强碱 is fully **dissociated** 解离 in water (almost every molecule splits into ions).
- a **weak acid** 弱酸 or **weak base** 弱碱 is only partly dissociated (most molecules stay whole).

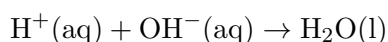
The **pH** scale measures how acidic a solution is: pure water is pH 7, acids are below 7, and alkalis are above 7. A strong acid has a lower pH than a weak acid of the same concentration.

You can tell strong and weak acids apart by:

- **reaction with a reactive metal**: a strong acid fizzes faster.
- **pH**: measured with a pH meter or **universal indicator** 通用指示剂.
- **electrical conductivity**: a strong acid conducts better, because it has more ions.

Neutralisation, salts and titration curves

Neutralisation 中和 happens when the hydrogen ions from an acid react with the hydroxide ions from an alkali:



A **salt** 盐 is also formed, from the rest of the acid and base.

In a **titration** 滴定 you add one solution to another and follow the pH. The **titration curve** 滴定曲线 has a steep, almost vertical jump near the end point. The exact shape depends on whether each reactant is strong or weak.

To pick an **indicator** 指示剂, choose one whose colour change falls inside that steep jump. For a strong acid with a strong base most indicators work; for a weak acid with a strong base you need one that changes in the higher pH range.